

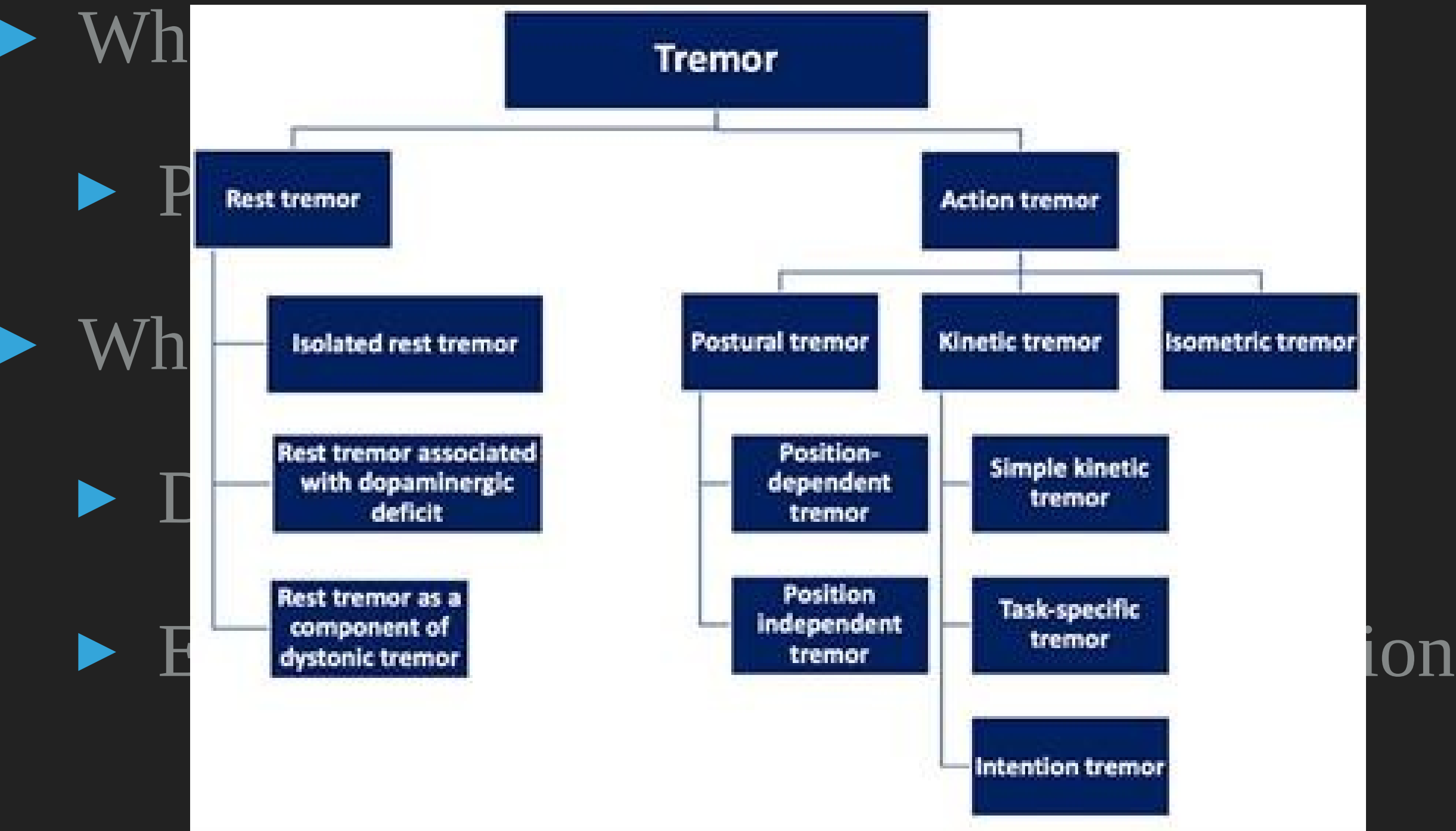
RESTING TREMOR SUPPRESSION DEVICE

EE300W CAPSTONE PROJECT

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INTRODUCTION

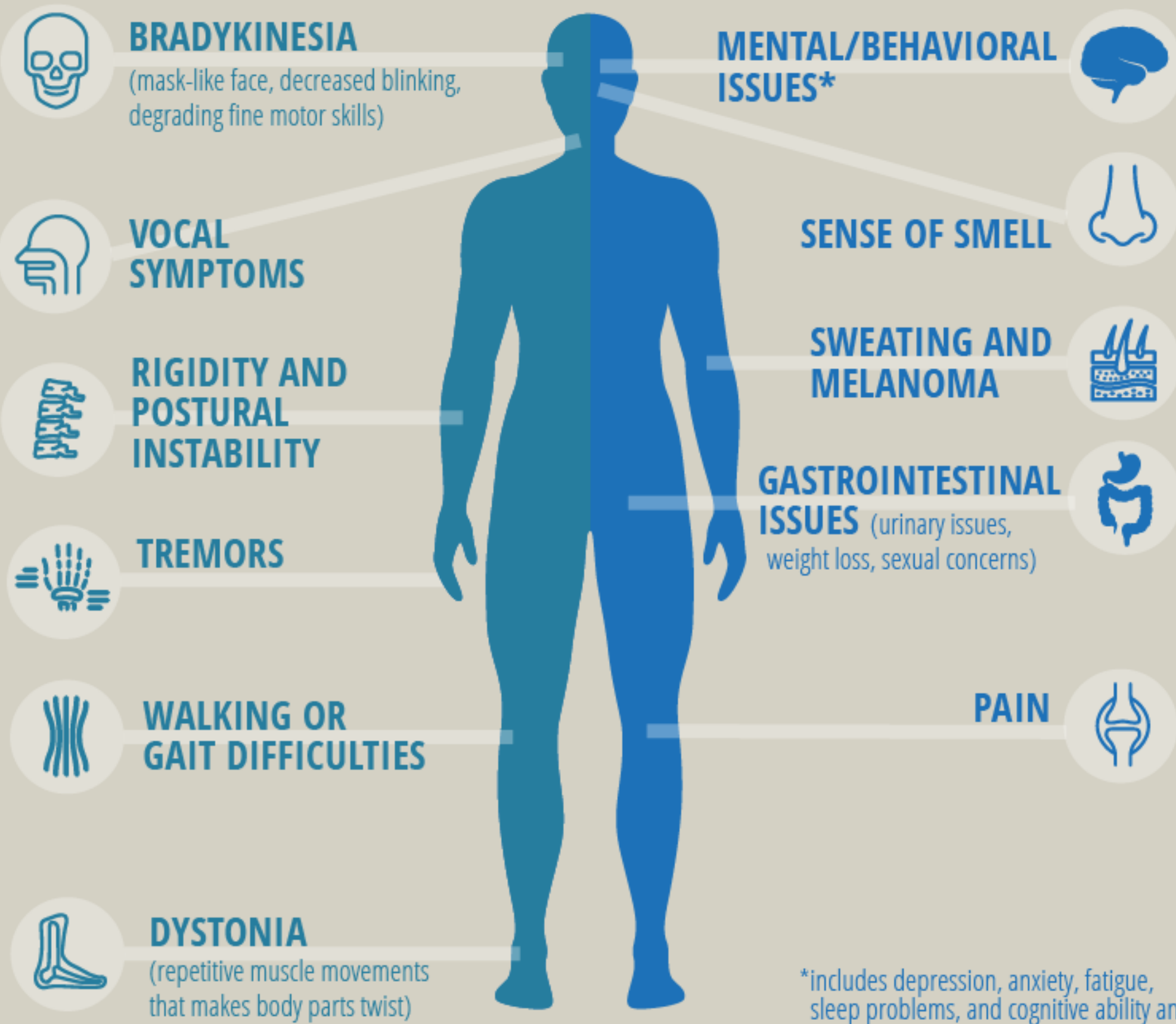


Rest tremor vs Action tremor *frontiersin*

PARKINSON'S DISEASE

Motor Skill Symptoms

Nonmotor Skill Symptoms

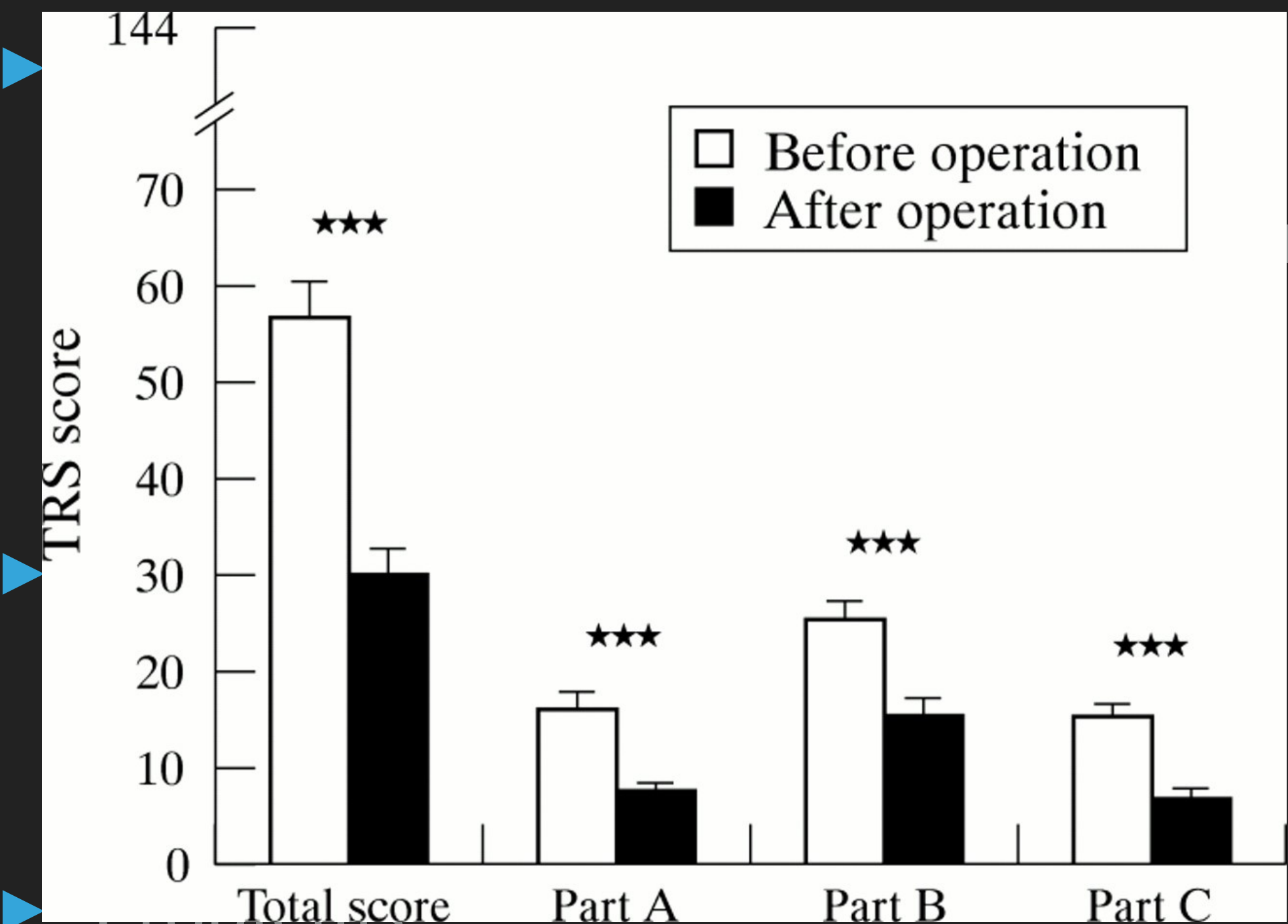


P.D. symptoms *pcori*

TREATMENT OPTIONS

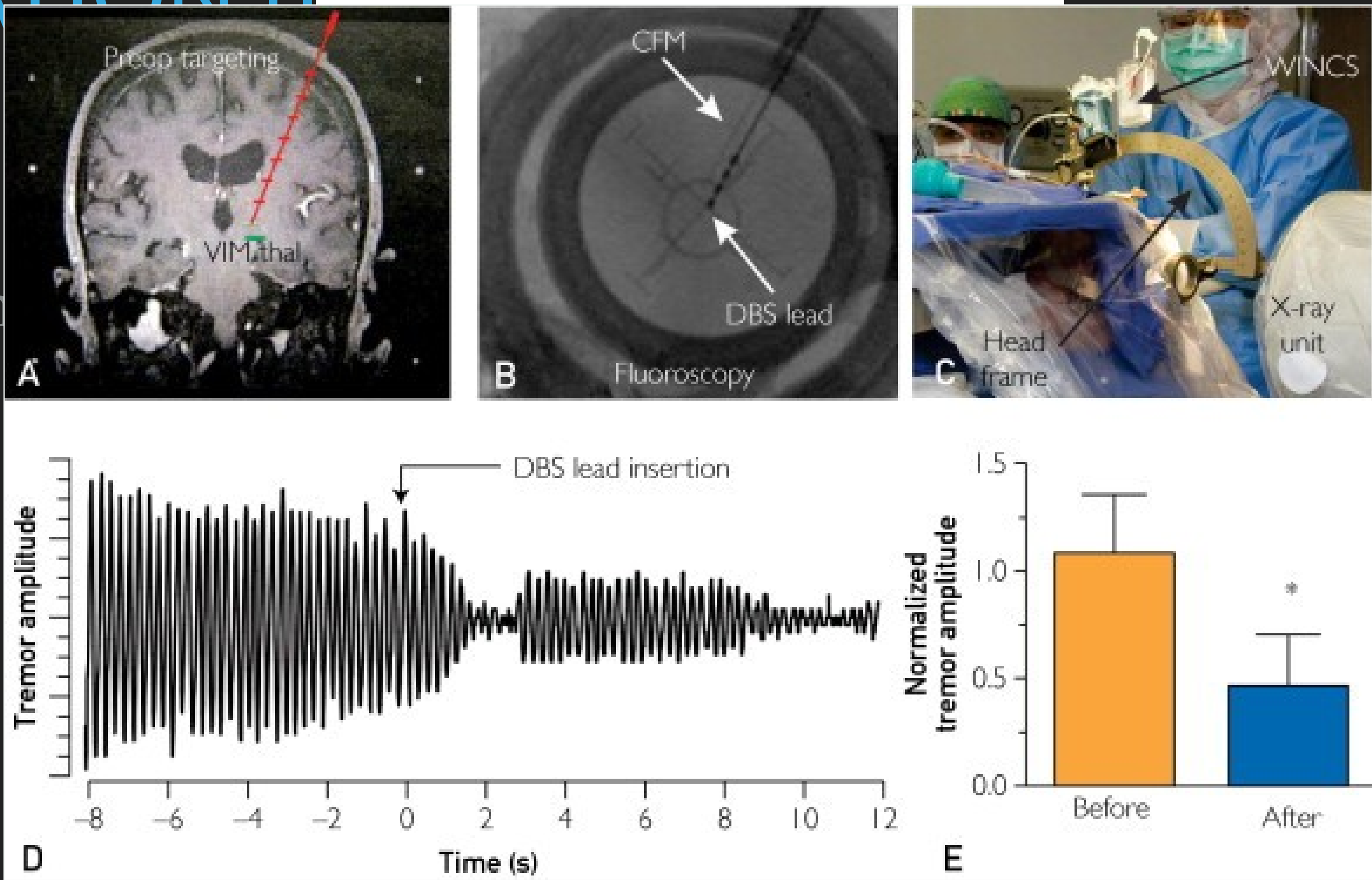
- ▶ Deep Brain Stimulation (D.B.S.)
- ▶ Allopathic Medication
- ▶ Neuromuscular Electrical Stimulation
- ▶ Passive Suppression Orthoses

DEEP BRAIN STIMULATION (DBS)



Tremor score before vs after D.B.S. procedure
bmj

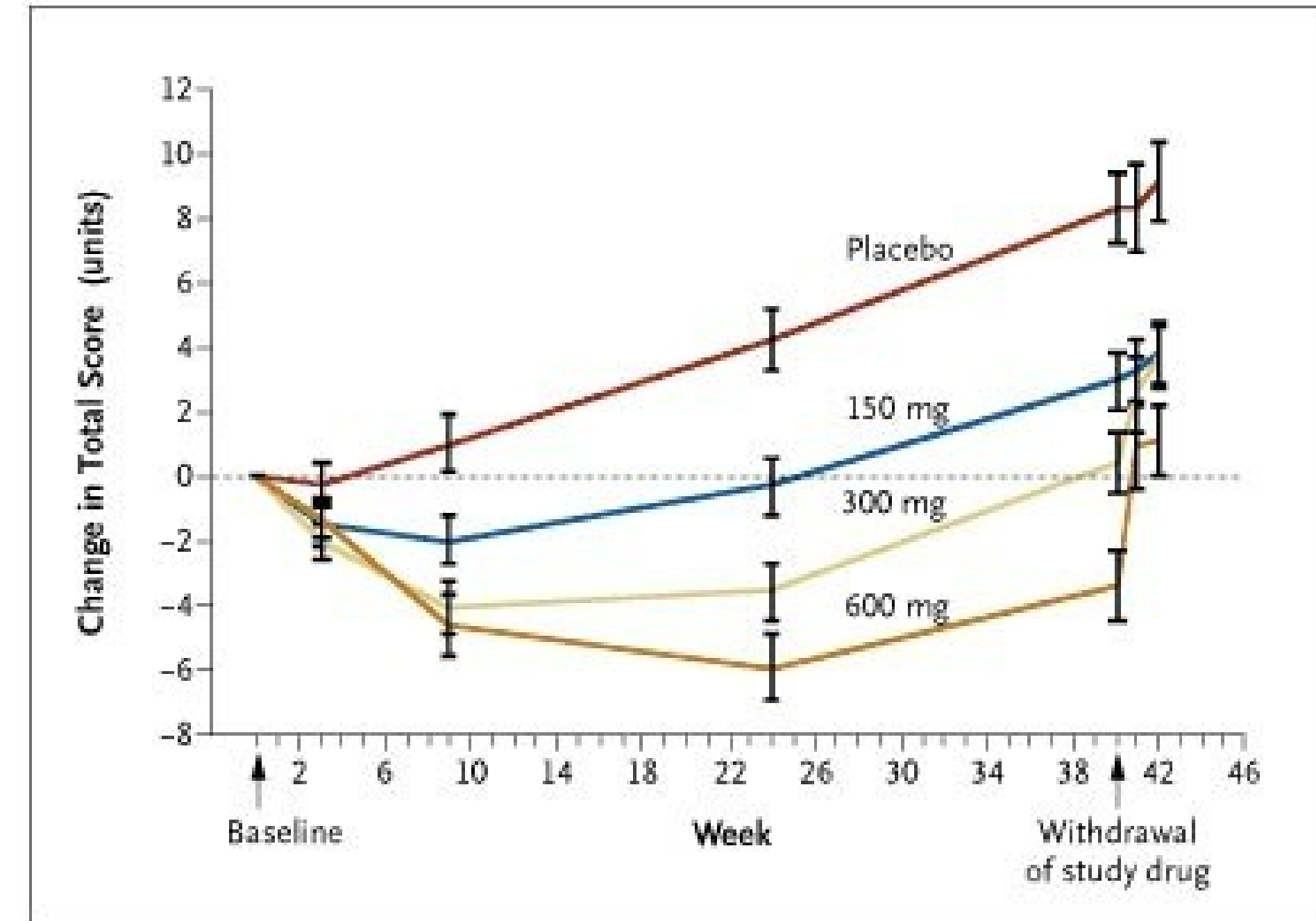
► Invasive surgery required



D.B.S. procedure and tremor amplitude graphs
elsevierhealth
parkinsonsNSW

ALLOPATHIC MEDICATION

- ▶ Mechanism:
 - ▶ Facilitate the production of Dopamine
 - ▶ Prevent the reuptake of Dopamine
- ▶ Results:
 - ▶ Significant reduction of symptoms
- ▶ Problems:
 - ▶ Regular re-administration of drugs needed
 - ▶ Possible side effects and drug interactions



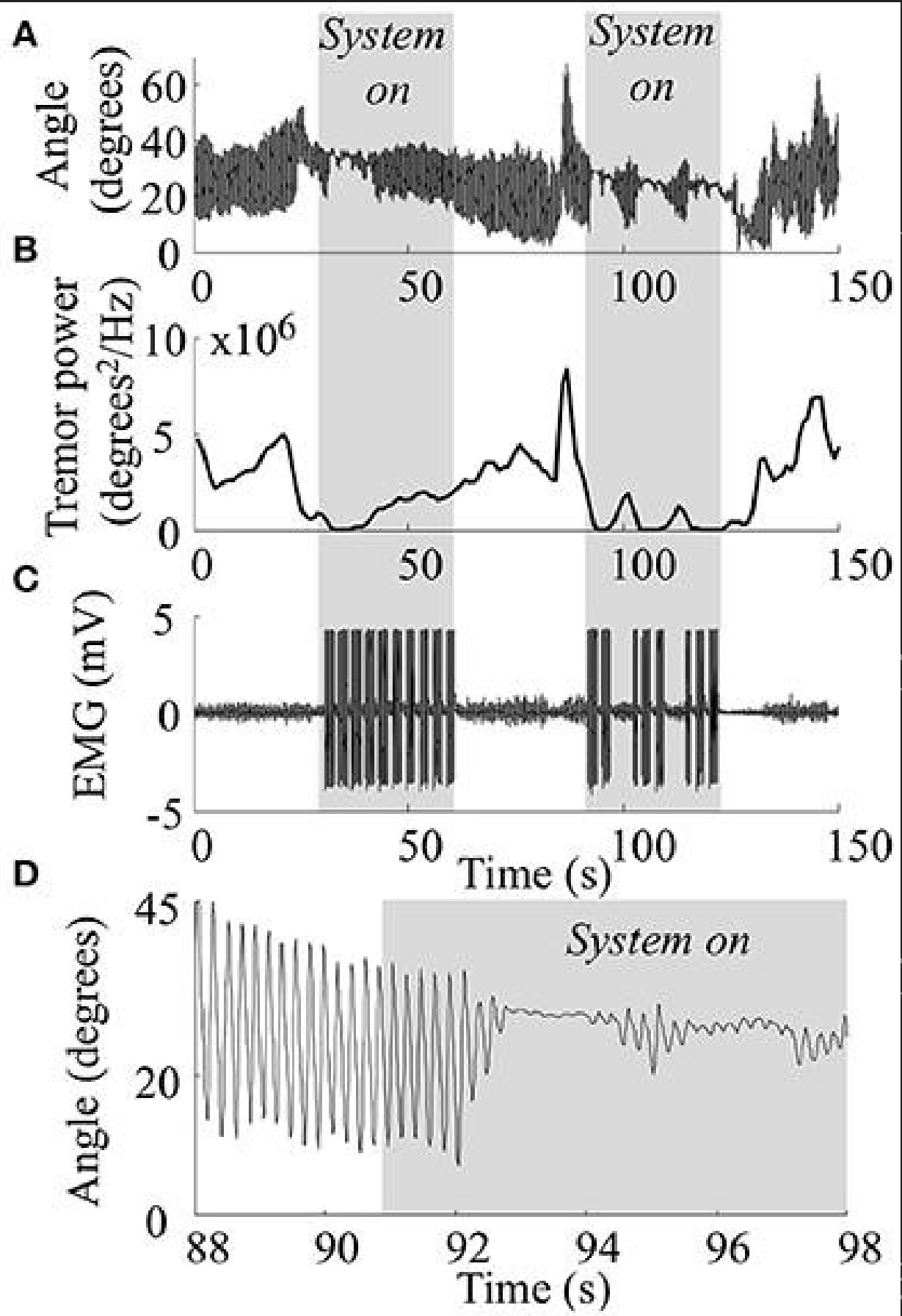
Base cause of P.D. everyday health

Change in tremor score vs medication dose *nejm*

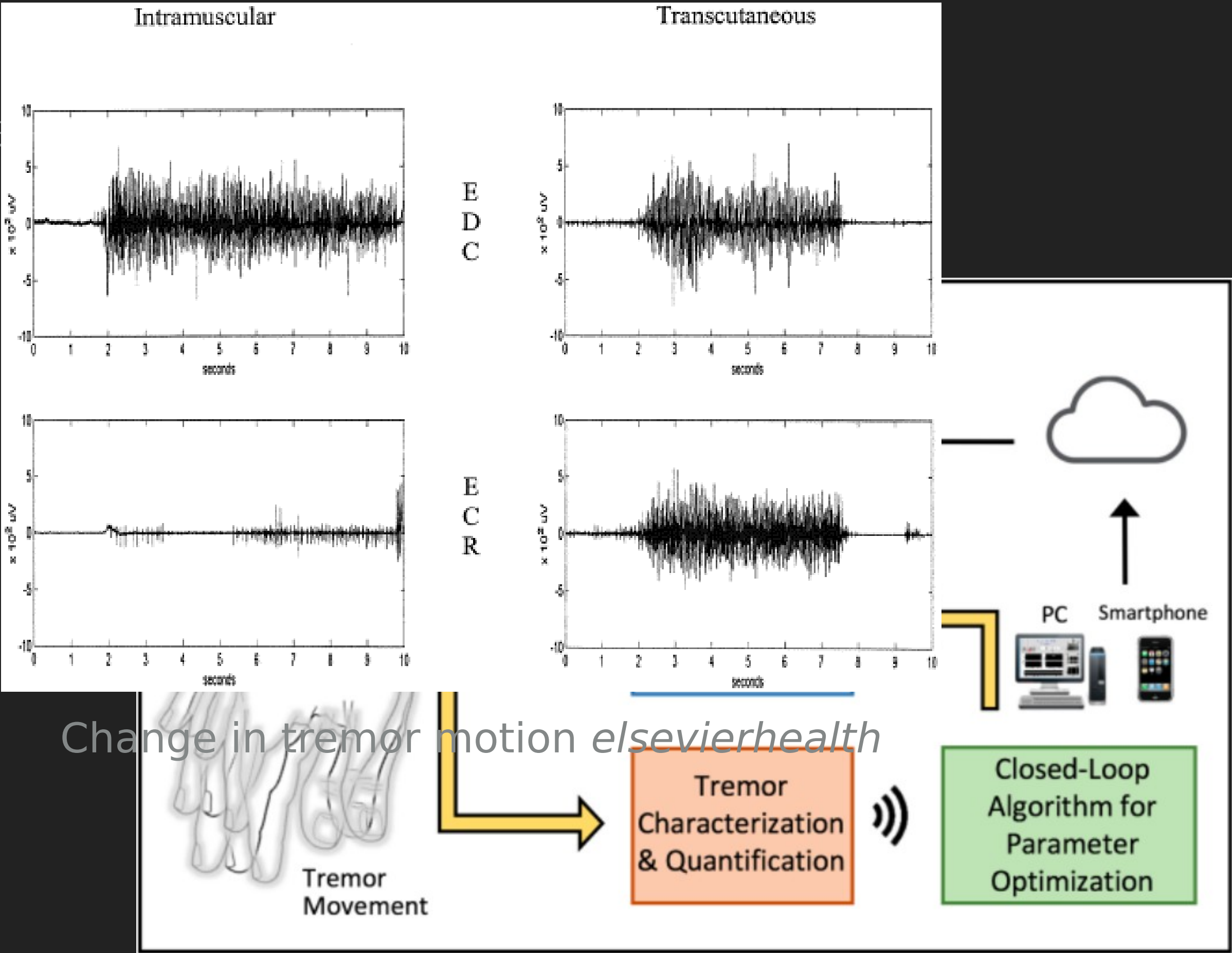
Working mechanism of P.D. medication *stemcellthailand*

NEUROMUSCULAR ELECTRICAL STIMULATION

- Types:
- Primary
- Mechanism
- Trigger
- Tremor
- Results:
- Moderate
- Problems
- Placem



Change in tremor characteristics *frontiersin*



PASSIVE SUPPRESSION ORTHOSES

- ▶ Rely on gyroscopic stabilisation or restrictive gear to reduce tremor magnitude
- ▶ Problems:
 - ▶ Bulky and cumbersome



GyroGlove *reutersmedia*



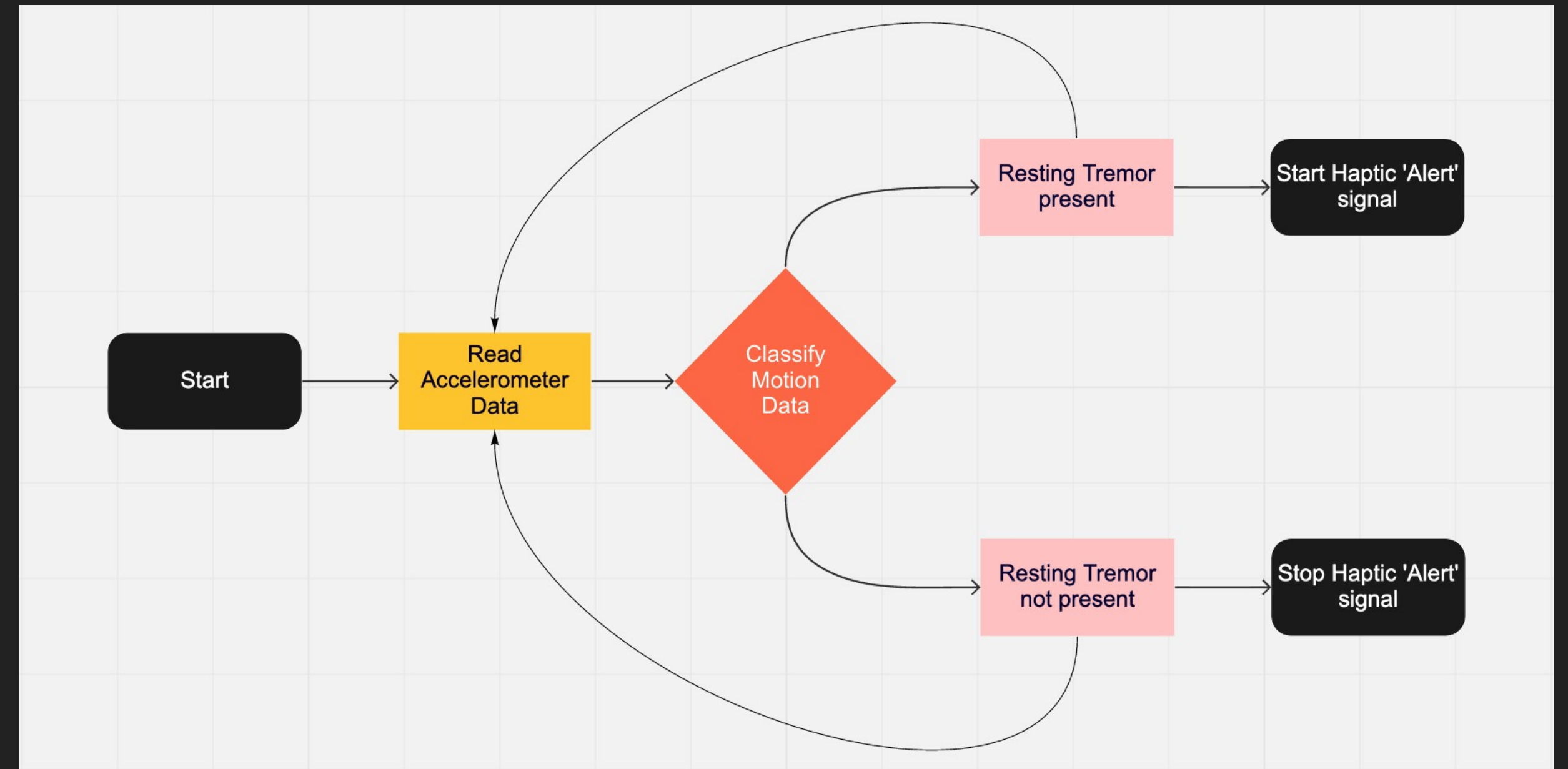
Steady-one glove *readi-steady*

SCOPE AND GOALS

- ▶ Suppression of resting tremor in P.D. patients
- ▶ Use sensor data to determine motion type
- ▶ Perform all computation off-line and in real-time
- ▶ User performs voluntary movement upon receiving feedback

SELECTED APPROACH

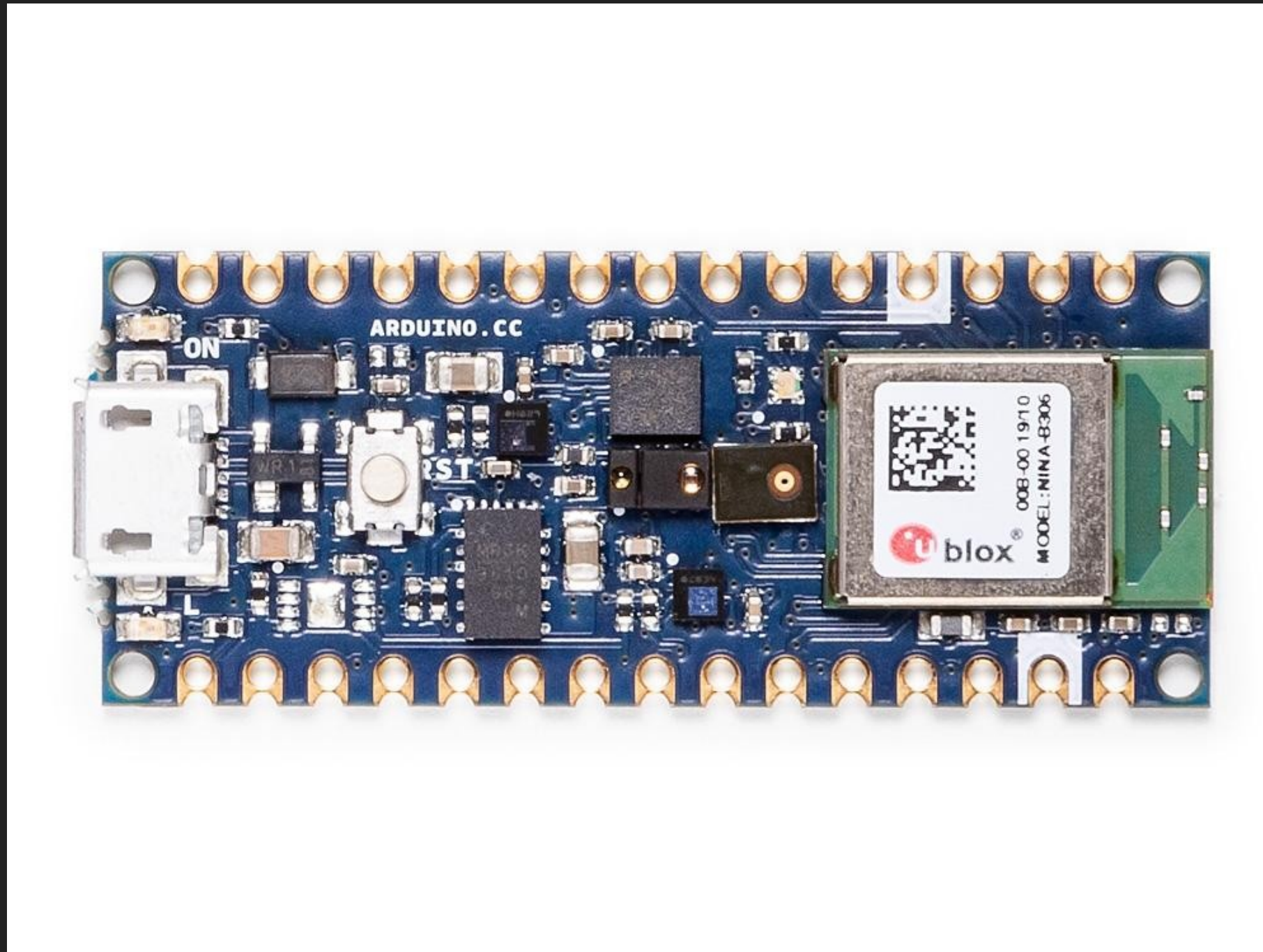
- ▶ ‘Brains’ of the project
- ▶ Collection and processing of data
- ▶ Classification of data to motion type
- ▶ Feedback system for user



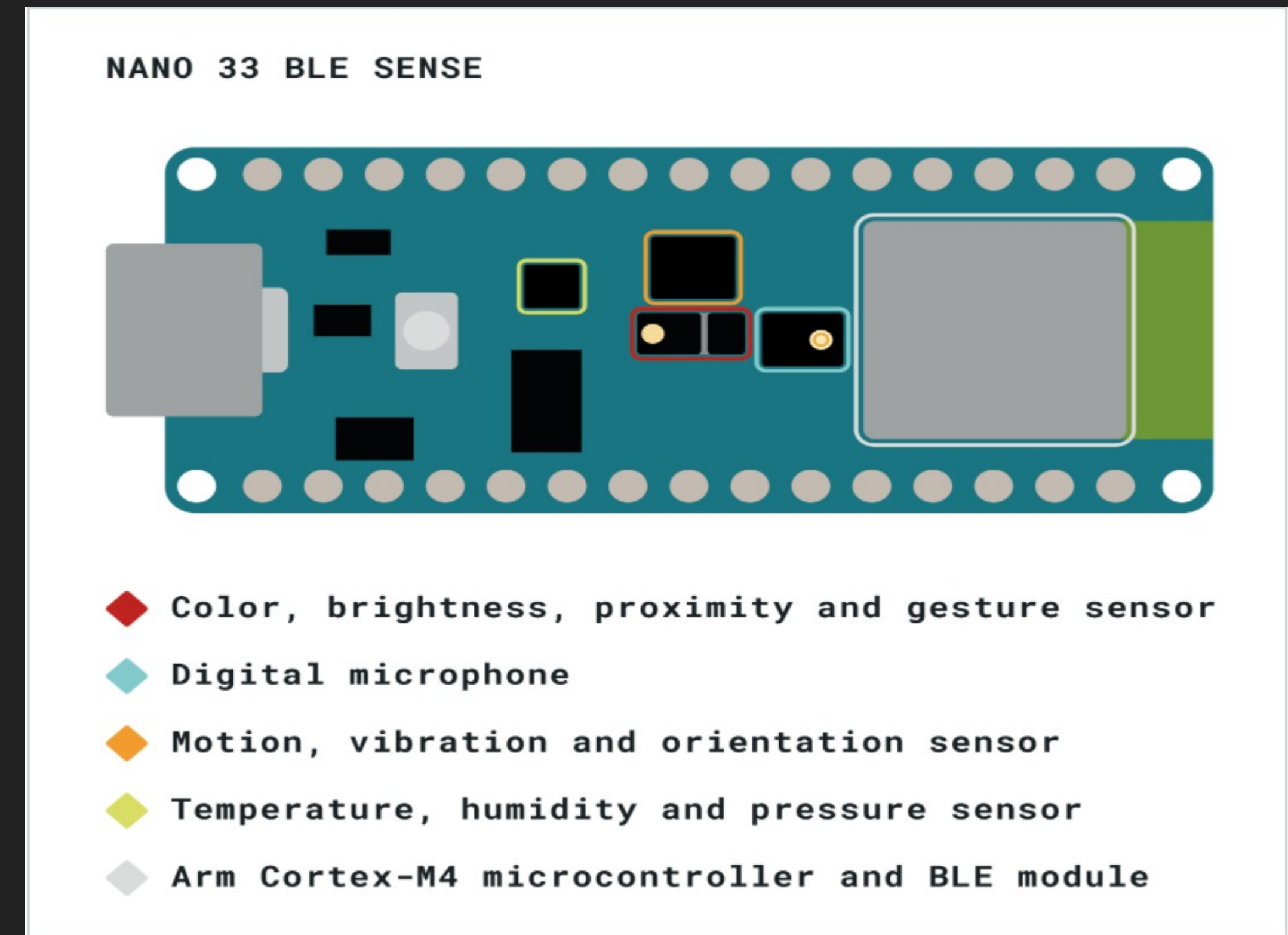
High level diagram of computation

‘BRAINS’ OF THE PROJECT

► Arduino Nano 33 BLE Sense



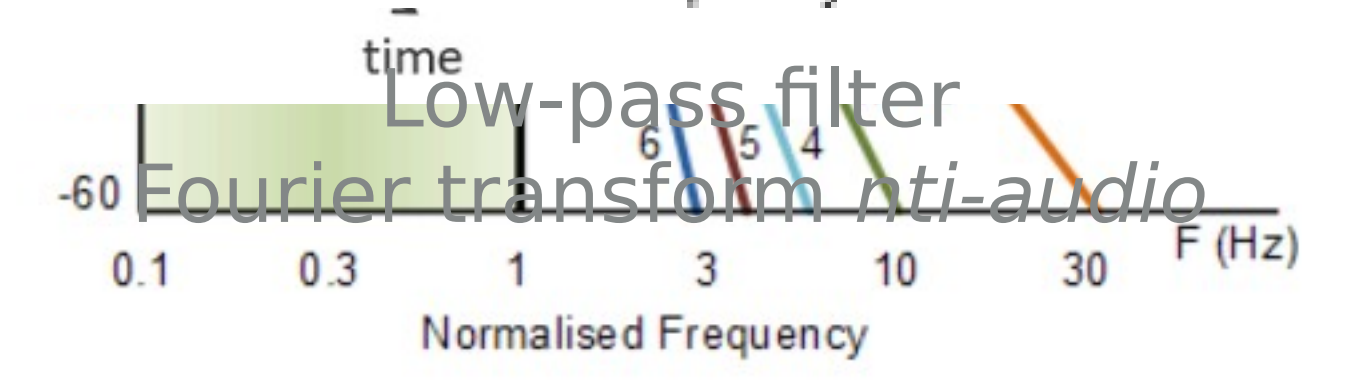
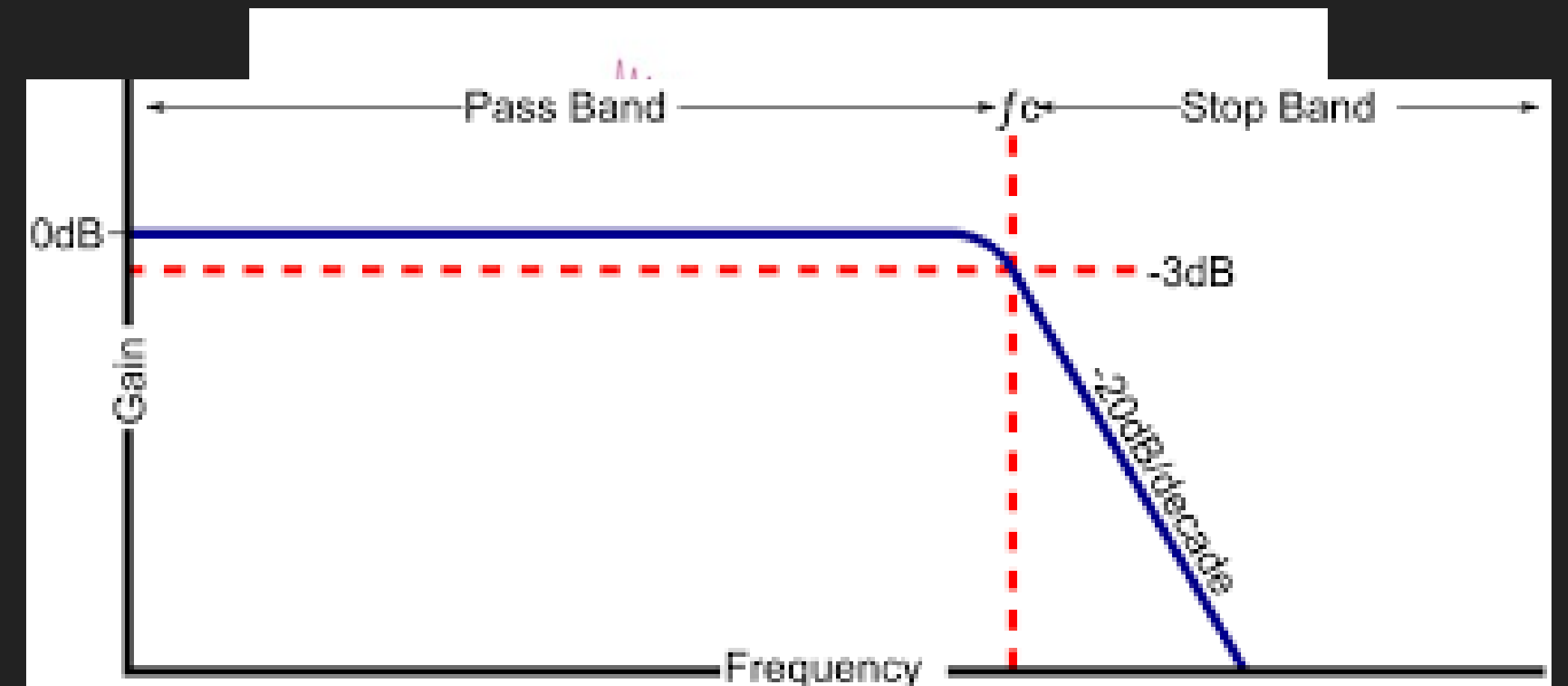
Arduino Nano 33 BLE Sense *arduino(2020)*



Component placement *arduino(2020)*

PROCESSING OF DATA

- ▶ Splitting data into fixed windows
- ▶ 6th order Butterworth filter
- ▶ Fourier Transform
- ▶ Frequency domain and spectral power features extracted

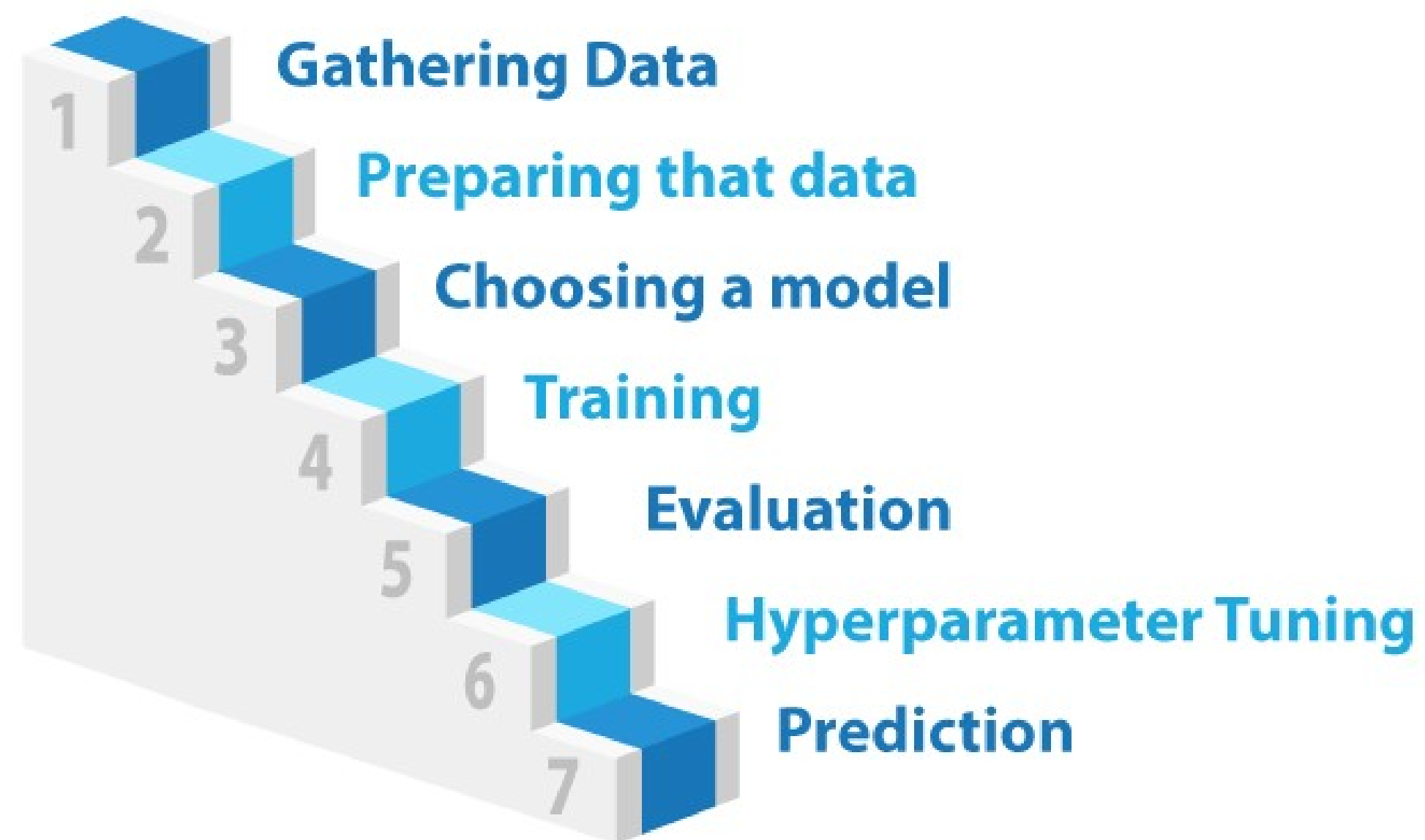


Butterworth filter *imgur*

CLASSIFICATION OF DATA TO MOTION TYPE

- ▶ What are machine learning
- ▶ Neural Networks
- ▶ Enables

7 steps of Machine Learning



K Keras

(TASK-DRIVEN)

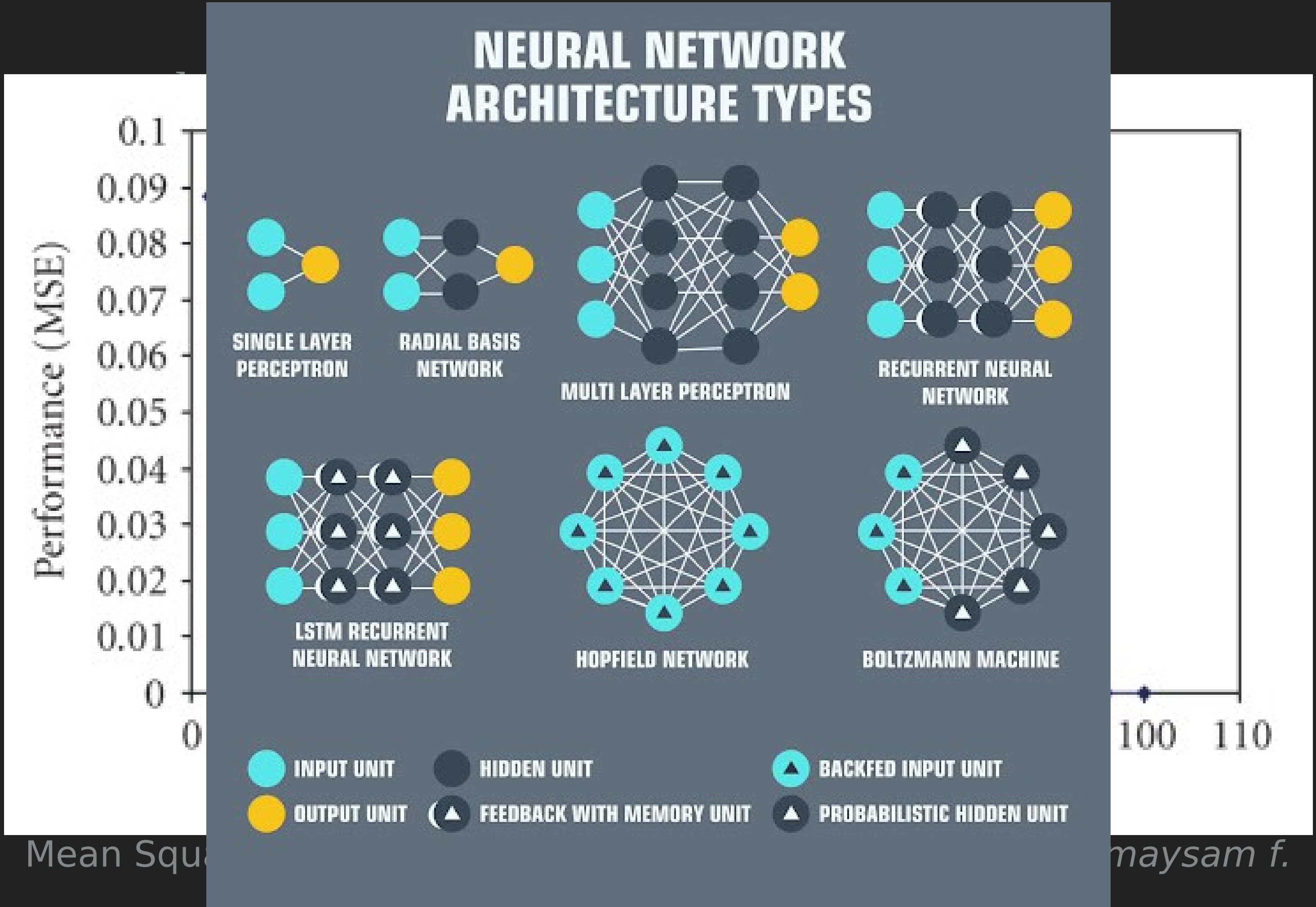
(DATA-DRIVEN)

(LEARN FROM ERRORS)

cloudfront
amazonaws

CHOOSING THE BEST STRUCTURE FOR THE NEURAL NETWORK

- ▶ Types of neural
- ▶ How to choose
- ▶ How to choose

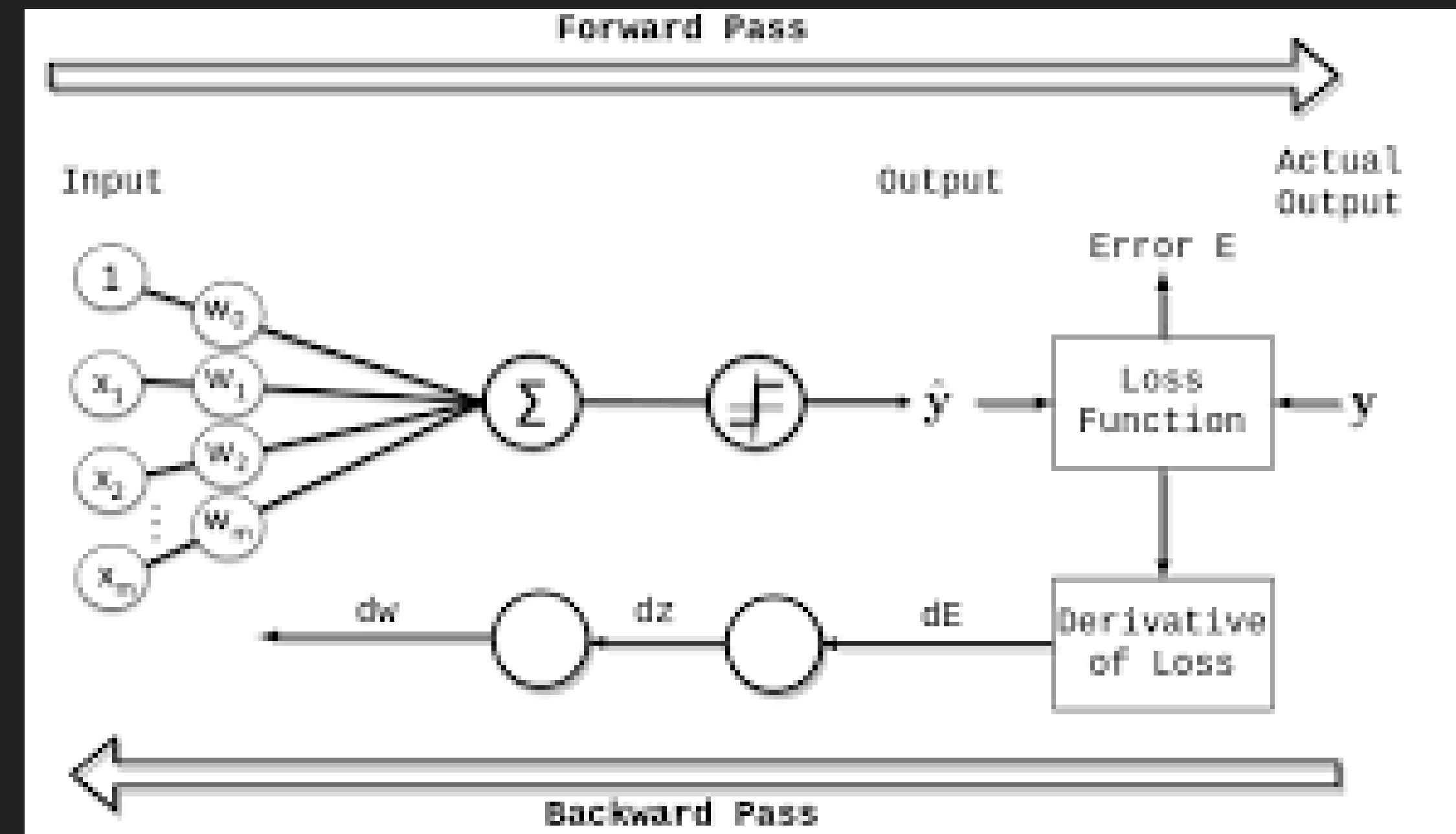


TRAINING OF THE NEURAL NETWORK

- ▶ How is a neural network trained?
- ▶ Number of training epochs?



Number of training epochs vs. error%



Single training epoch

RESULTS

- ▶ Accuracy: 90%
- ▶ Loss: 0.650

On-device performance ?



INFERENCIN...
1 ms.



PEAK RAM U...
1.7K



FLASH USAGE
19.0K

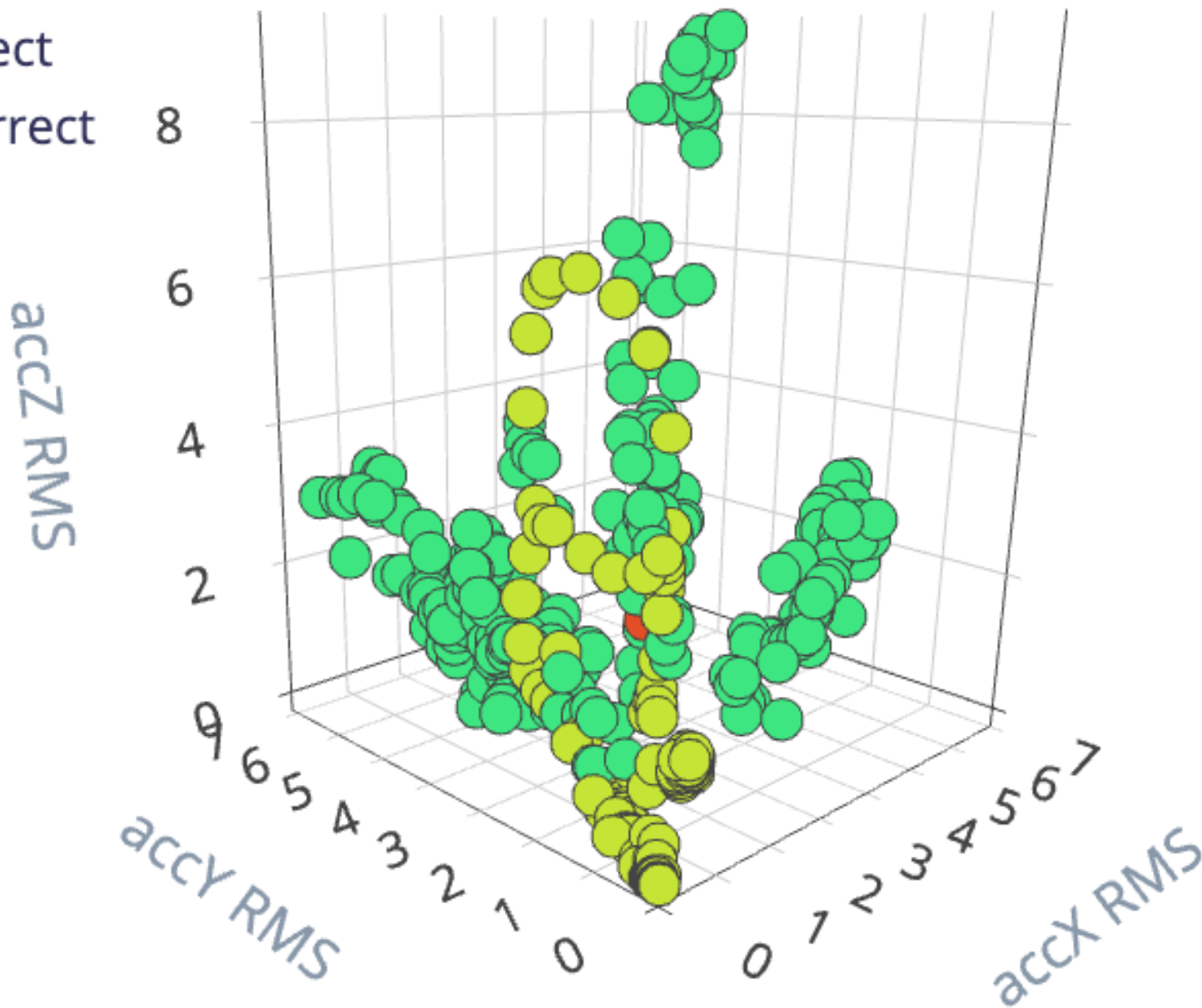
Feature explorer (full training set) ?

accX RMS ▼

accY RMS ▼

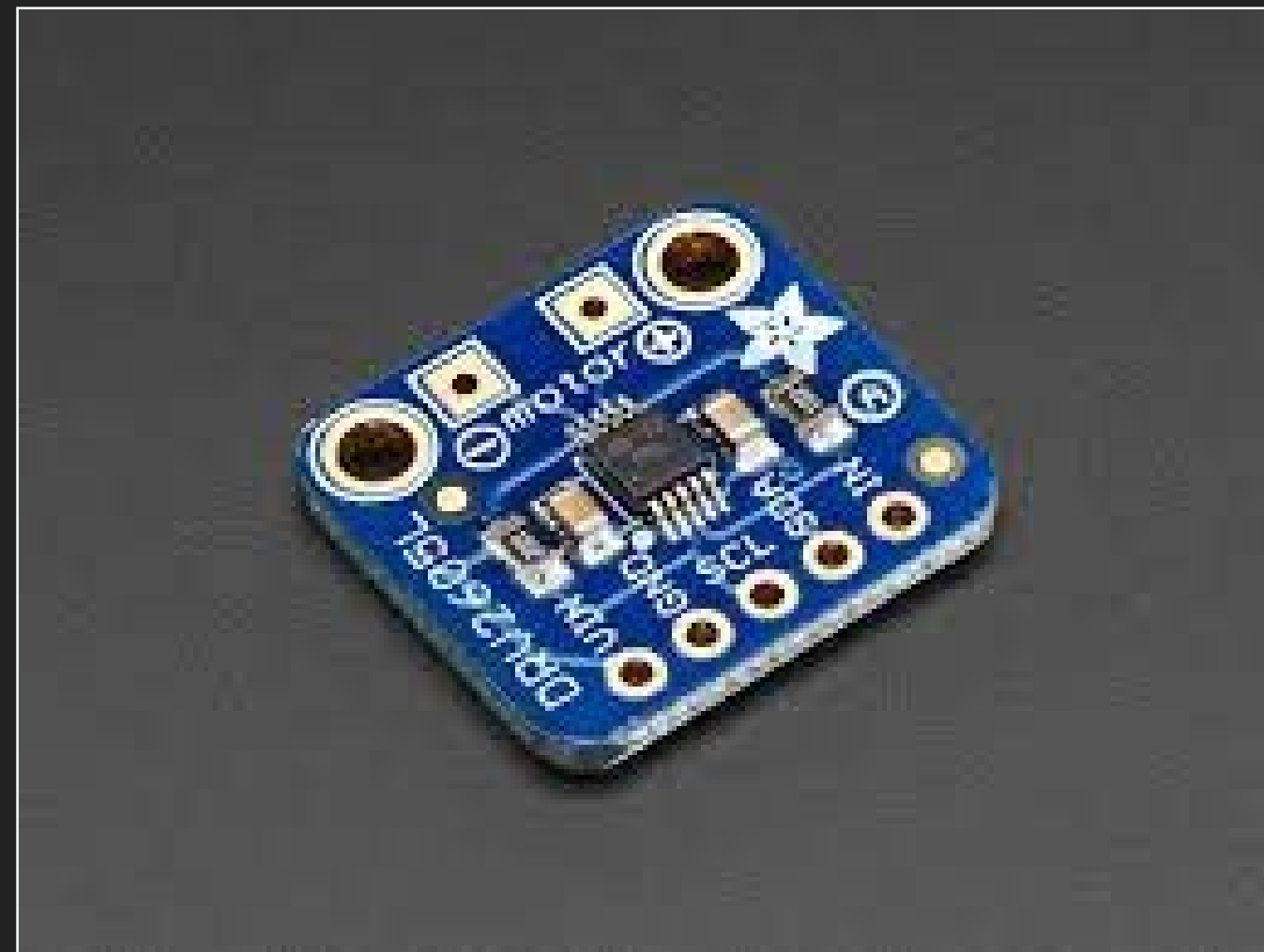
accZ RMS ▼

- idle - correct
- updown - correct
- updown - incorrect

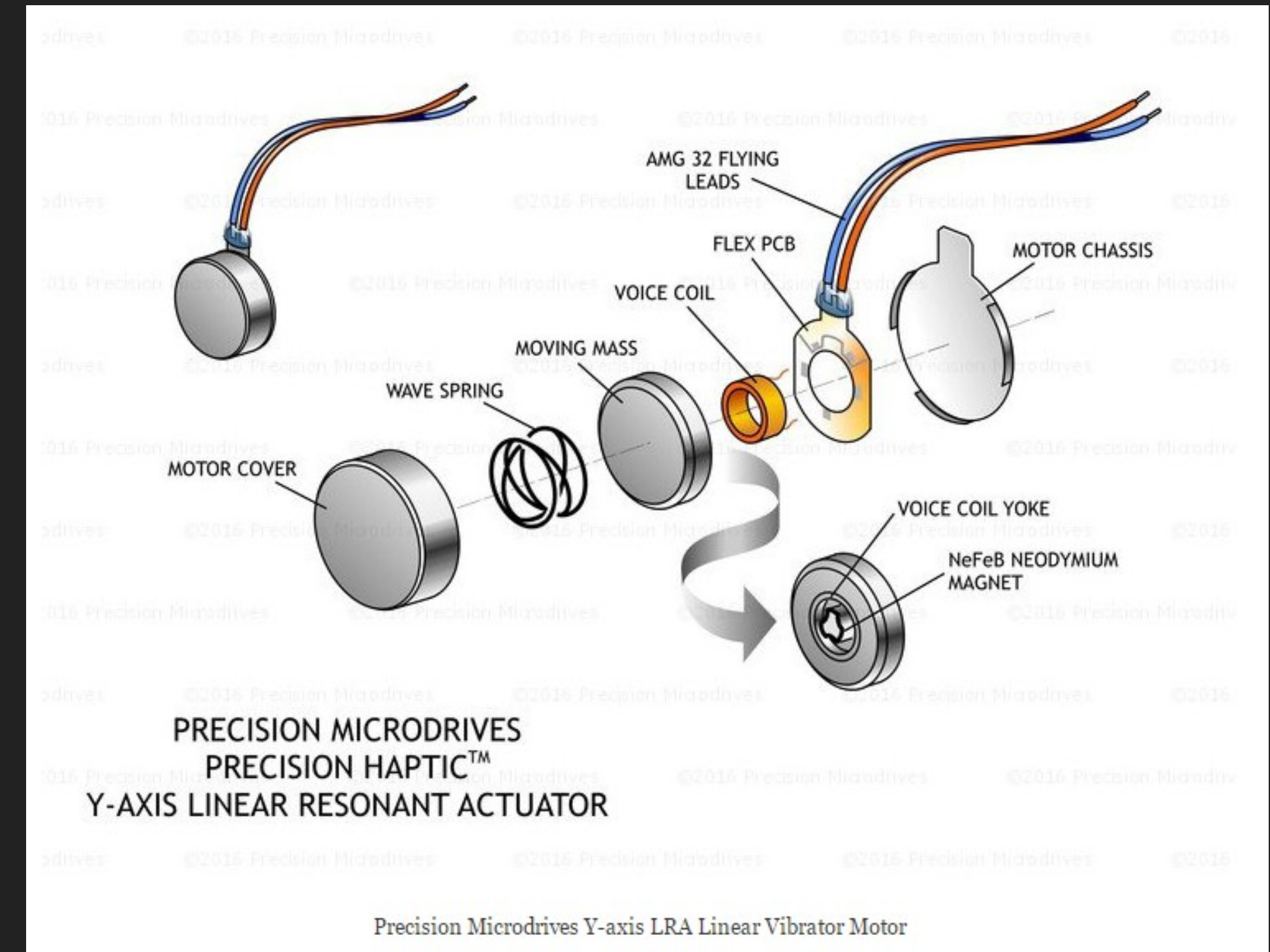


FEEDBACK MECHANISM

- ▶ Haptic feedback using DRV2605L and ERM motor
- ▶ Waveform generation for the motor

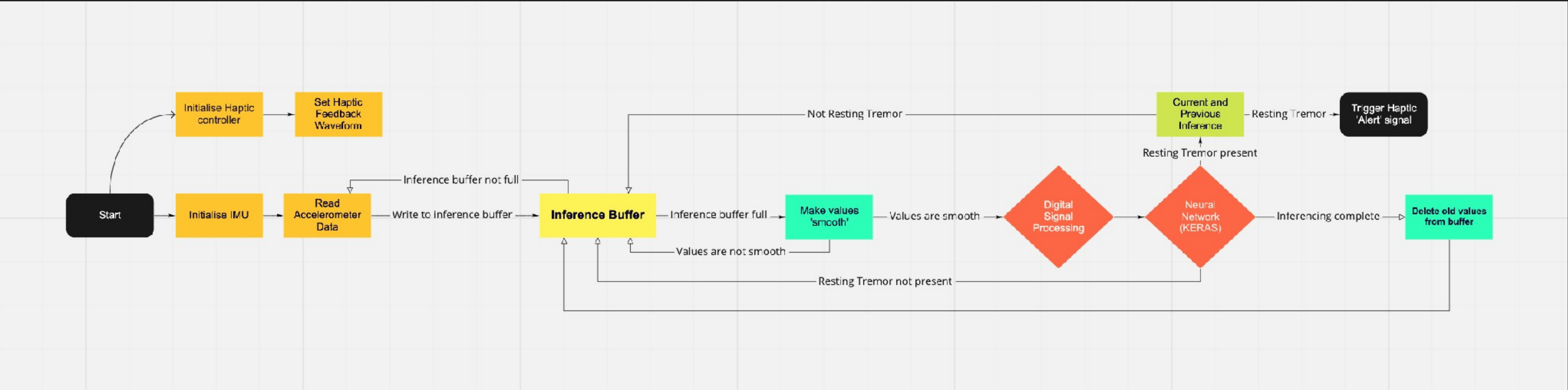


DRV2605L Haptic Motor
Controller *adafruit*



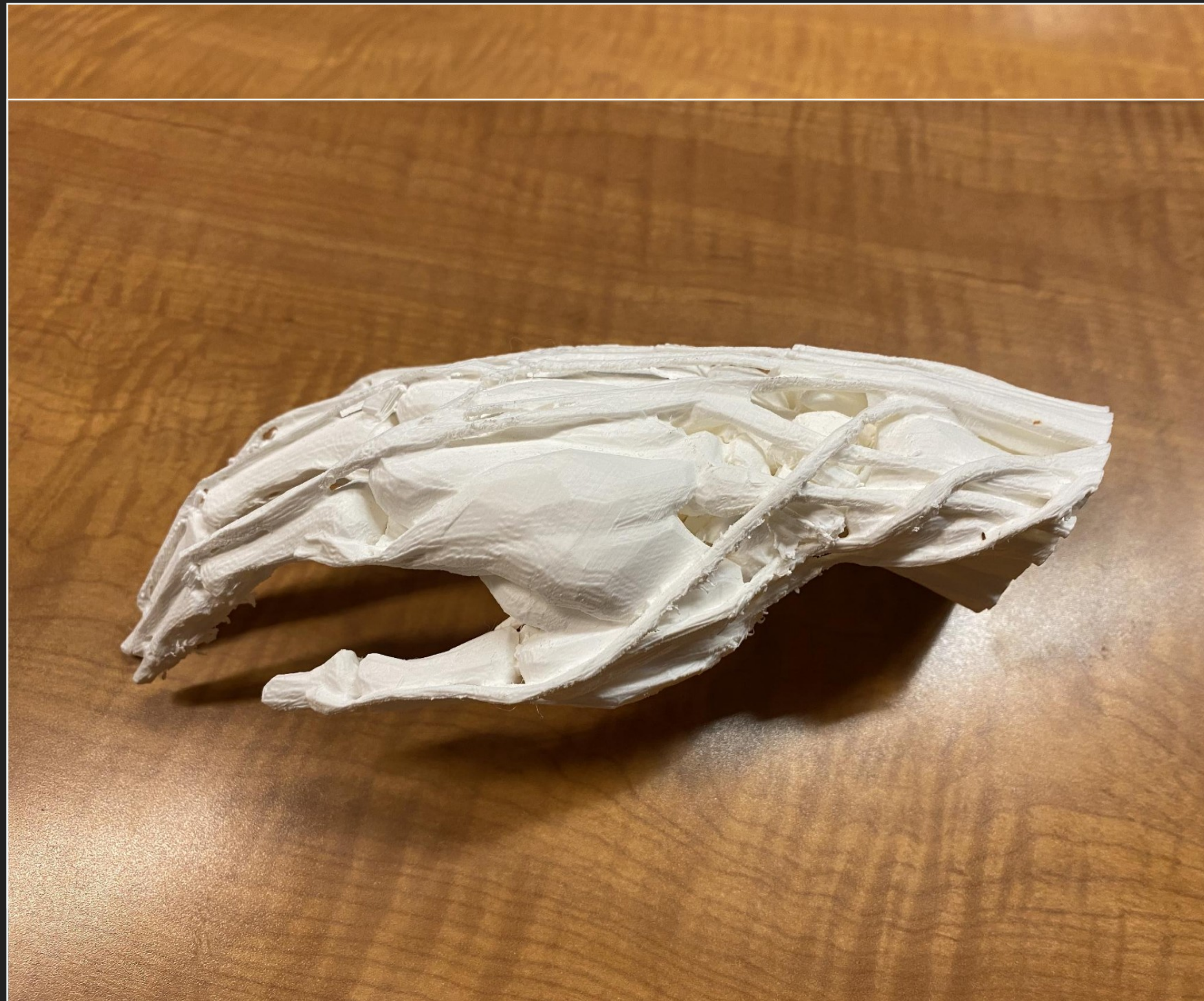
Eccentric Rotating Mass Motor Construction
sparkfun

LOW LEVEL DIAGRAM

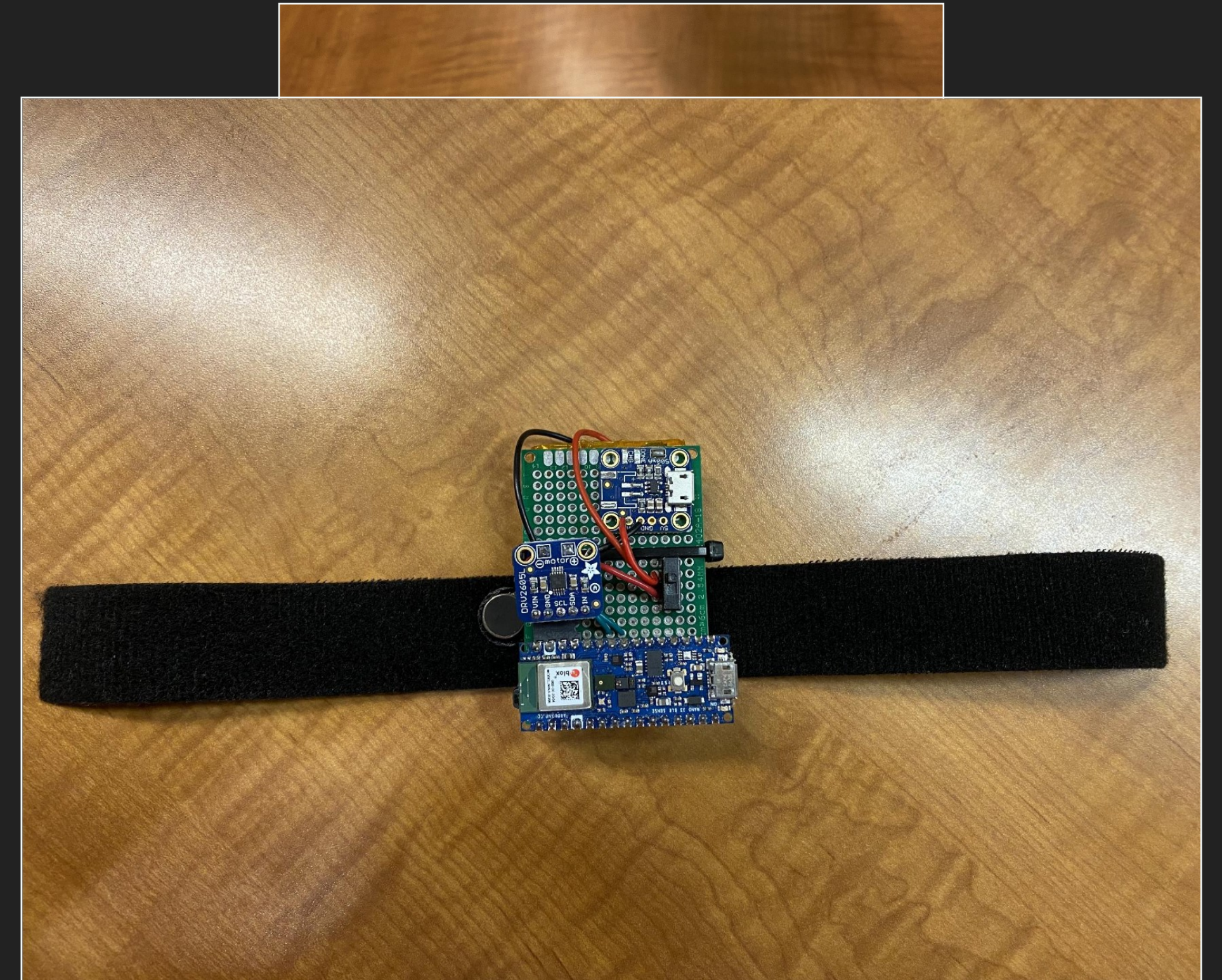


Low level diagram of computation

PROTOTYPE



Human hand model with peripheral nerves
and tendons



Prototype device

FUTURE PLANS

- ▶ Distinguish between Essential tremor and resting tremor in P.D. patients
 - ▶ Added suppression of Essential tremor
- ▶ Assign UPDRS scores to classifications
- ▶ Tracking disease progression
- ▶ BLE updates to phone
- ▶ Upload collected data to cloud for future re-training

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